2.1.6 To test the equality of two means – paired *t*-test

Procedure:

Step 1 : Let X and Y be two correlated random variables having the distributions respectively $N(\mu_X, \sigma_X^2)$ (Population-1) and $N(\mu_Y, \sigma_Y^2)$ (Population-2). Let $D = X - Y$, then it has normal distribution $N(\mu_D = \mu_X - \mu_Y, \sigma_D^2)$.

Frame null hypothesis as

$H_0 : \mu_D = 0$

And choose alternative hypothesis from

(i) $H_1 : \mu_D \neq 0$ (ii) $H_1 : \mu_D > 0$ (iii) $H_1 : \mu_D < 0$

Step 2 : Describe the sample/data. Let $(X_1, X_2, \ldots, X_m)$ be a random sample of m observations drawn from Population-1 and $(Y_1, Y_2, \ldots, Y_n)$ be a random sample of n observations drawn from Population-2. Here, these two samples are correlated in pairs.

Step 3 : Set up level of significance (α)

Step 4 : Consider the test statistic

$$T = \frac{\bar{D}}{\frac{S}{\sqrt{n}}} \text{ under } H_0 .$$

where $\bar{D} = \frac{\sum_{i=1}^{n} D_i}{n}$; $D_i = X_i - Y_i$ and $S = \sqrt{\frac{\sum_{i=1}^{n} (D_i - \bar{D})^2}{n-1}}$.

The approximate sampling distribution of the test statistic T under H_0 is t - distribution with $(n-1)$ degrees of freedom.

Step 5 : Calculate the value of T for the given data as

$$t_0 = \frac{\bar{d}}{\frac{s}{\sqrt{n}}}$$

where $\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$; $d_i = x_i - y_i$ (sample mean) and

$s = \sqrt{\frac{\sum_{i=1}^{n} (d_i - \bar{d})^2}{n-1}}$ (sample standard deviation)

Step 6 : Choose the critical value, t_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu_D \neq 0$	$\mu_D > 0$	$\mu_D < 0$
Critical Value (t_e)	$t_{n-1, \frac{\alpha}{2}}$	$t_{n-1, \alpha}$	$-t_{n-1, \alpha}$

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